

Adapttime: Method Overview

Thesis project

Current protocol

Abstract

Adapttime studies whether a small fitted adaptor can combine a time-series foundation model with fixed, calendar-aligned retrieval while preserving the official TIME test protocol. This note defines the current univariate `full_ridge_shared` formulation; it reports no empirical result.

1 Chronological protocol

Each dataset and forecast term is divided into a fixed datastore, adaptation training, adaptation validation, and the untouched official TIME test interval. The first three regions end strictly before the test. Ridge candidates are fit once on adaptation training, K and α are selected on adaptation validation, and the selected coefficients are frozen for test. There is no train-plus-validation refit.

For a query time t , let $d_{\max}$ be the fixed datastore endpoint and P the dataset period. The latest eligible neighbor date is

$$d_t = \min(t - H, d_{\max}) - (\min(t - H, d_{\max}) - t) \bmod P. \quad (1)$$

Earlier candidates walk backward from d_t by a datastore stride that is a multiple of P . This keeps the datastore fixed while aligning its phase to the query.

2 Full shared ridge

For each query, V is the vanilla foundation forecast and C is the forecast obtained when the K retrieved neighbor trajectories are supplied as covariates. For neighbor i , Y_i is its observed horizon and N_i is the foundation model’s vanilla forecast from that neighbor’s own history. The full design is

$$X = [V, C, Y_1, \dots, Y_K, N_1, \dots, N_K]. \quad (2)$$

The paired Y_i and N_i terms expose neighbor residual information. After normalizing each window and channel by its query-context standard deviation, one no-intercept ridge vector is shared across all forecast horizons:

$$\hat{\beta}_\alpha = \arg \min_{\beta} \|(Y - V) - X\beta\|_2^2 + \alpha\|\beta\|_2^2, \quad (3)$$

$$V' = V + X\hat{\beta}_\alpha. \quad (4)$$

The primary setting is $K = 10$, $\alpha = 10^{-2}$. Validation searches $K \in \{1, 5, 10, 15\}$ and $\alpha \in \{10^{-3}, 10^{-2}, 10^{-1}\}$. Delta, convex, per-horizon, and multivariate Adapttime variants are outside the current protocol.

3 Evaluation

The official test compares vanilla V , retrieval-context C at the selected K , and frozen Adapttime V' . It retains per-window and per-user MSE, MAE, normalized MSE, and normalized MAE, then reports equal-window and equal-user means and population standard deviations. The TIME-wide view selects only completed task manifests, averages exact repeats before optionally averaging distinct scientific configurations, averages terms within each dataset/frequency, and then gives each dataset/frequency equal weight. Conclusions require completed and inspected cluster artifacts.